

Structured summary: latent-class and mixture-model formulations for biomarker-treatment interaction in N-of-1 trials (supplementary key-points summary)

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2026-08-12

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This document is a structured, section-by-section summary of the key points argued in the companion manuscript, “Latent-class and mixture-model formulations for biomarker-treatment interaction in N-of-1 trials.” It is provided as a supplementary reading aid and does not stand on its own; all notation, derivations, simulation detail, and citations are in the main manuscript.

1 Introduction

1.1 The aggregated N-of-1 trial and the biomarker-moderation question

The aggregated N-of-1 design pools within-person treatment contrasts across participants for population-level inference.

- Each participant cycles between active treatment and control across multiple within-person blocks.
- The design outperforms parallel-group trials in information per participant when treatment response is reversible on a useful timescale.
- Deployed in chronic pain, neuropsychiatric PTSD, and rare-disease settings where parallel-group trials are infeasible.

The primary scientific estimand in biomarker-guided N-of-1 trials is the biomarker-treatment interaction term.

- A predictive biomarker enables point-of-care treatment stratification per the PATH framework.
- Power depends on moderation magnitude, within-participant contrast precision, trial design, and carryover leakage.
- Carryover erodes the cleanness of within-subject contrasts and is the principal nuisance parameter in power calculations.

The DGP encoding of biomarker moderation determines carryover-induced power loss, a question the prior literature raised but did not resolve.

- Power loss under carryover varies sharply with how moderation is encoded in the DGP.
- Two formal answers from outside biostatistics – continuous heterogeneity and latent-class structure – dominate the available options.
- This paper maps those traditions and their consequences for N-of-1 trial design and analysis.

1.2 Two formal answers: continuous heterogeneity and latent-class structure

The continuous-heterogeneity answer encodes biomarker moderation as a smooth interaction in a single multivariate distribution.

- Every participant has some treatment response; the response is a linear function of the biomarker.
- Population heterogeneity is captured by a single distribution with an interaction parameter $\beta_{bm:D}$.
- This is the dominant tradition in clinical biostatistics via random slopes and fixed-effects interaction terms.

The latent-class answer encodes biomarker moderation as discrete subpopulation structure, with the biomarker predicting class membership.

- Participants belong to unobserved responder/non-responder classes with qualitatively distinct treatment-response curves.
- Class membership is a Bernoulli draw with biomarker-dependent mixing probabilities.
- Marginal moderation emerges from the mixture, not a smooth conditional-mean function.

The two encodings reflect genuinely distinct biological hypotheses that diverge under the longitudinal structure of an N-of-1 trial.

- Continuous heterogeneity implies a graded physiological substrate scaling drug effect; latent-class implies qualitatively distinct phenotypes.
- Both views yield identical marginal associations at a single timepoint but diverge once carryover and serial dependence are engaged.
- Methodology from outside biostatistics – with mature formalization and identifiability analysis – is directly applicable.

1.3 Psychometric origins

The continuous-trait tradition traces to Spearman and the factor-analytic lineage of Thurstone and Lawley, in which heterogeneity is by assumption continuous.

- Observed indicators are modeled as linear combinations of unobserved continuous factors plus residual noise.

- Between-person heterogeneity is summarized by factor scores along one or more dimensions.
- This tradition forms the psychometric foundation for the continuous-heterogeneity encoding of biomarker moderation.

The latent-class tradition originates with Lazarsfeld’s local-independence formulation and was placed on rigorous ML footing by Goodman, then extended to longitudinal and covariate-dependent forms.

- Local independence within classes is the founding assumption of latent class analysis (LCA); Goodman derived identifiability conditions and ML estimation.
- Extensions through the 1970s–1980s cover polytomous and continuous indicators, multilevel data, and covariate-dependent class probabilities.
- McLachlan and Peel’s monograph is the standard entry point for finite mixture models.

The psychometric literature has never resolved the continuous-versus-discrete question, motivating hybrid factor mixture models.

- Bartholomew argued the data rarely settle the continuous-discrete choice without strong prior assumptions.
- Cognitive ability favors continuous factors; certain psychopathologies favor discrete subtypes.
- Yung and Arminger developed factor mixture models combining discrete classes with within-class continuous factors.

1.4 Econometric and marketing-science development

Heckman and Singer’s econometric work on unobserved heterogeneity anticipated finite-mixture methodology, warning that structural estimates are sensitive to the assumed heterogeneity distribution.

- The econometric concern was unobserved heterogeneity in micro-econometric models where identical-agent assumptions fail.
- Heckman and Singer proposed non-parametric ML with the heterogeneity distribution estimated as a discrete mixture.
- Lindsay unified the theory of mixture likelihoods; this work directly informed biostatistical finite-mixture practice.

The marketing-science literature formalized covariate-dependent gating functions and clusterwise regression, providing the vocabulary now used in mixture-of-experts models.

- DeSarbo and Cron introduced clusterwise linear regression with cluster-specific coefficients; Wedel and DeSarbo extended it to GLMs.
- The gating function – class-membership probability as a function of covariates – became a first-class object.

- Jacobs et al. formalized the same structure as 'mixtures of experts' in machine learning, supplying the standard vocabulary.

Two econometric themes translate directly to the N-of-1 trial setting: identifiability concerns at small N , and biomarker-as-gating-covariate structure.

- Econometric identifiability concerns are acute when heterogeneity is the principal inference target rather than a nuisance.
- The biomarker-treatment moderation question is whether the gating function depends on the biomarker and within-class slopes differ.
- These two themes directly motivate the present paper's framing.

1.5 Convergence on hybrid frameworks

Psychometric and econometric traditions converged on factor mixture models combining discrete class structure with within-class continuous heterogeneity.

- Yung developed the factor mixture model (FMM) as a finite mixture of confirmatory factor models with class-specific loadings and residual covariances.
- Arminger placed FMMs in the Mplus structural equation framework.
- Lubke and Muthen provided the first systematic Monte Carlo evaluation of conditions under which class structure is distinguishable from continuous heterogeneity.

Growth mixture models and group-based trajectory models extend the finite-mixture framework to longitudinal outcome trajectories.

- Growth mixture models (GMMs) posit class-specific trajectories with within-class random-effects variation; group-based models constrain within-class variance to zero.
- Nagin's group-based approach originated in criminology and is now widely used in epidemiology and clinical research.
- Both families are the longitudinal analogues of the latent-class DGP developed in the present paper.

Overextraction hazards tempered early optimism: GMMs can spuriously identify classes from non-gaussian single-component data, and reliable class recovery requires large samples.

- Bauer and Curran showed GMMs extract apparent classes from non-gaussian single-component DGPs.
- Lubke and Muthen documented analogous FMM hazards and identified joint conditions on N , class separation, and parameter differences.
- Nylund et al. systematised class enumeration via Monte Carlo evaluation of information criteria and the bootstrap likelihood-ratio test (BLRT); the cumulative message is that hundreds to thousands of participants are required.

1.6 Translation into longitudinal biostatistics

Verbeke and Lesaffre introduced finite-mixture random-effects distributions into biostatistics via longitudinal HIV data, with Zhang and Davidian extending to semi-parametric specifications.

- The model augments a standard random-intercept-and-slope linear mixed model (LMM) with a discrete latent class indicator on the random-effects population.
- Their motivating application was qualitatively distinct CD4 trajectories in HIV-infected patient latent classes.
- Zhang and Davidian allowed more flexible heterogeneity distributions via semi-parametric mixture specifications.

The `lcmm` and `flexmix` R packages translate finite-mixture longitudinal models into accessible software for biostatistical applications.

- `lcmm` implements joint latent-class mixed models for longitudinal outcomes with optional joint survival modeling.
- `flexmix` provides a general regression-mixture framework with user-definable component models including mixed-effects specifications.
- Both have been used in pharmacokinetic, cognitive aging, and clinical latent-class discovery applications.

Mixture and latent-class methods have not been systematically applied to aggregated N-of-1 designs; the biomarker-moderation question has been encoded exclusively as a continuous interaction.

- Published N-of-1 methods focus on within-participant estimation, Bayesian hierarchical models, and meta-analytic aggregation.
- Biomarker-moderation power calculations use continuous interaction terms with no parallel latent-class formulation.
- The mixture-formulation question appears unaddressed in the existing N-of-1 methodology literature.

1.7 Why this matters for N-of-1 designs specifically

Under a latent-class DGP, class membership is fixed across drug states and is in principle recoverable from the full trajectory even when on-drug contrasts are noisy.

- Continuous-heterogeneity DGPs encode moderation in how the within-participant contrast varies with the biomarker.
- Latent-class DGPs encode moderation in class membership, which is invariant to drug state.
- The two encodings imply different efficient analysis strategies and different power profiles under carryover.

N-of-1 trial sample sizes are well below the threshold at which mixture-model identifiability has been validated, so a misspecified single-component analysis may be the more reliable tool.

- Psychometric identifiability validation requires hundreds to thousands of participants; N-of-1 trials operate at $N \in [30, 150]$.
- Even a biologically correct latent-class DGP may yield underpowered or unidentified mixture estimates at trial-relevant N .
- The mixture-versus-mitigation trade-off is an empirical question not addressed at N-of-1-relevant sample sizes.

Carryover erodes the moderation signal directly under continuous-heterogeneity but only indirectly under latent-class, predicting divergent power-loss profiles.

- Under continuous heterogeneity, carryover attenuates $D_{bc,it}$ and thereby directly attenuates the moderation signal.
- Under latent-class structure, carryover erodes within-class drug contrasts but does not directly attenuate class membership.
- The two formulations predict different power-loss profiles; this difference is empirically large in the companion carryover- sensitivity work.

1.8 Aims and structure of the paper

The paper has four aims: formalize latent-class DGPs for N-of-1 trials, characterize estimation strategies at small N , benchmark power via simulation, and re-analyze the prazosin-PTSD dataset.

- Aims 1–2 are methodological: formal specification and estimation/diagnostic characterization with explicit identifiability conditions.
- Aim 3 is computational: a factorial simulation benchmarking mixture-DGP against continuous-DGP power under carryover.
- Aim 4 is applied: re-analysis of the Hendrickson et al. (2020) prazosin-PTSD data under the latent-class formulation.

The paper moves from notation and DGP specification through multivariate normal (MVN) approximation, identifiability analysis, model taxonomy, and spectrum organization to analysis implications and a discussion of open questions.

- Sections on the faithful DGP, MVN approximation, and failure regimes develop the theoretical core.
- The finite-mixture taxonomy and biomarker-moderation spectrum organize the model landscape systematically.
- The discussion addresses three open questions and situates the companion five-study simulation program.

2 Notation

Core notation: B_i is the biomarker, BR_{it} the biological-response component, $D_{bc,it}$ the continuous-decay drug indicator, and Z_i the unobserved class label.

- Outcome is decomposed into BR_{it} , TV_{it} , and PB_{it} following the Hendrickson framework.
- $D_{bc,it} = \exp(-\lambda t_{sd,it})$ off-drug and 1 on-drug; $\lambda = \ln 2/t_{1/2}$.
- The DGP and analysis model are permitted to use different carryover half-lives.

3 Faithful latent-class data-generating processes

The faithful latent-class DGP is a finite mixture with biomarker-dependent class-membership probability and class-specific modGompertz drug-response distributions.

- $Z_i \mid B_i \sim \text{Bernoulli}(\pi(B_i))$ with $\pi(B_i)$ typically a logistic function of the biomarker.
- Class-conditional distributions f_0 and f_1 differ in drug-response parameters (peak response, onset rate, or both).
- Within-class temporal structure uses the same modGompertz form as single-class formulations.

Class-label invariance preserves moderation information across carryover that the MVN-covariance encoding, where the signal is entirely in the second moment, cannot retain.

- Z_i is fixed across drug states; it carries participant-level moderation information even when between-class mean separation is attenuated.
- Under a full carryover off-drug period, class-conditional means converge but Z_i remains recoverable from the full trajectory.
- Under MVN-covariance encoding the carryover erosion of the covariance is direct and no residual signal survives the off-drug period.

4 MVN approximation to mixture-induced moderation

The achievable marginal c_{bm} under an MVN approximation is bounded by the geometric mean of the two between-class variance fractions.

- c_{bm}^2 equals the product of the between-class variance fractions in B and in BR .
- Small class separation in either variable limits the achievable marginal c_{bm} even when class structure is otherwise well-defined.
- The MVN approximation reproduces the mixture's first two moments under within-class independence.

Within the regularity conditions, the MVN second-moment approximation is adequate for LMM power calculations and captures differential correlation across drug states.

- The approximation yields elevated correlation on-drug and shrinkage toward zero off-drug after carryover has decayed.
- Adequacy requires adequate class overlap and π bounded away from 0 and 1.
- For t -tests on $\beta_{bm:D}$ this approximation captures covariance consequences without committing to a discrete DGP.

5 Implications for power under carryover

The power gap between a second-moment analysis and a mixture-faithful analysis upper-bounds what a class-aware procedure can recover, but the bound is asymptotic and may reverse at small N .

- The 40–60% covariance-only power loss reflects direct erosion of a second-moment signal; class membership is carryover-invariant.
- The gap between covariance-only and mixture-faithful power estimates upper-bounds the recoverable power.
- At finite N in the trial-relevant range, finite-sample bias can drive the realized gap below the bound or reverse its sign.

The Study 1 factorial crosses two DGPs and two analyses to decompose approximation error from analysis-model mis-specification and to verify Type I error control.

- The mixture-DGP/MVN-analysis vs. mixture-DGP/mixture-analysis contrast quantifies power recoverable by a faithful analysis.
- The two MVN-analysis cells quantify approximation error in the second-moment encoding.
- A null check on the MVN-DGP/mixture-analysis cell verifies no spurious power extraction.

6 Identifiability at trial-relevant sample sizes

Mixture EM requires either informative class-probability priors or strong class separation; neither is assured at N-of-1 sample sizes, which fall well below identifiability validation thresholds.

- Posterior entropy stabilization requires hundreds to thousands of participants in published psychometric applications.
- N-of-1 trials operate at $N \in [30, 150]$, well below this range.
- The identifiability cost is acute precisely when the moderation effect's existence is uncertain.

The operative question is not faithfulness but whether the fidelity gain is recoverable at realistic N , or whether standard analysis mitigations offer a better power-per-complexity trade.

- Mixture analysis may be underpowered or unidentified even under a genuinely mixture-structured DGP at trial-relevant N .

- Analysis-strategy mitigations within standard mixed-effects practice have better-understood inferential properties at small N .
- Study 5 of the companion simulation plan addresses this directly with small- N identifiability and Type I error checks.

7 Finite-mixture taxonomy

7.1 Latent class and latent profile analysis

LCA and latent profile analysis (LPA) are cross-sectional mixture models estimating class membership from observed indicators; in the trial context they characterize responder classes from response trajectories.

- LCA uses categorical indicators; LPA uses continuous indicators; both return posterior class-membership probabilities.
- In the biomarker-trial context, an LPA-style model estimates responder-class probabilities from on-drug and off-drug response trajectories.
- Classical estimation is via EM; standard R implementations include poLCA and Latent GOLD.

7.2 Growth mixture models and group-based trajectory models

GMMs extend LPA to longitudinal data with class-specific drug-response curves and within-class random-effects variation; group-based trajectory models constrain within-class variance to zero.

- A GMM is a longitudinal LMM with the random-effects population replaced by a mixture of multivariate normals.
- Group-based trajectory models are the degenerate case with zero within-class random-effect variance.
- Both families model longitudinal trajectory heterogeneity as the principal scientific question.

GMM is the closest analogue of the faithful DGP in Section 3, but Bauer and Curran’s overextraction cautions apply with full force at N-of-1 sample sizes.

- A biomarker-gated GMM is essentially the faithful latent-class DGP with drug-state-invariant class labels.
- GMM can extract apparent class structure from single-class non-gaussian data at realistic sample sizes.
- N-of-1 participant counts are typically well below those at which GMM class-count recovery has been validated.

7.3 Factor mixture models

Factor mixture models combine discrete class structure with within-class continuous factor heterogeneity, subsuming pure factor analysis

and pure LCA as limiting cases.

- Within-class variation is governed by a factor model; between-class variation by class-specific means, loadings, or residual covariances.
- The pure continuous (single-class factor) and pure discrete (LCA) cases are special cases of the FMM framework.
- FMMs match a wider range of biological scenarios at the cost of higher estimation requirements.

In FMM, the biomarker can function simultaneously as a noisy class indicator via the gating function and a continuous within-class modulator via factor loadings.

- The upper level assigns participants to classes as in LCA; the lower level gives each class its own random-effects structure.
- The biological hypothesis is distinct responder classes with additional continuous heterogeneity within each class.
- The hybrid representation is more biologically flexible than either pure LCA or pure factor analysis.

7.4 Regression mixtures and mixtures of experts

Regression mixtures replace a single regression surface with class-specific surfaces governed by a covariate-dependent gating function, paralleling propensity-score stratification in causal inference.

- Each class has its own regression surface; the gating model governs which surface applies to each participant.
- The biomarker enters both the gating function and (via class membership) the within-class regression coefficient on treatment.
- The two-stage structure parallels propensity-score stratification followed by within-stratum effect estimation.

In the trial simulation idiom, regression mixtures implement a DGP where the biomarker governs both responder probability and within-responder effect magnitude; `flexmix` provides the R implementation.

- The biomarker enters the gating function and determines within-responder drug effect magnitude.
- `flexmix` implements this family with user-definable component models including mixed-effects specifications.
- This family is suitable for repeated-measures DGP construction in aggregated N-of-1 simulation.

8 The biomarker-moderation spectrum

Locating the covariance-only formulation within the broader DGP spectrum clarifies which features of the carryover-induced power loss are properties of the specification rather than the science.

- The spectrum is organized by which moments the biomarker moderates: mean, covariance, both, or higher moments.
- Covariance-only is the Hendrickson formulation; mean-and- covariance and higher-moment forms are the generalizations.
- Understanding the spectrum is prerequisite to a complete sensitivity analysis of carryover-induced power loss.

8.1 Covariance-only moderation

The covariance-only specification fixes the conditional mean and encodes the interaction entirely in how strongly the biomarker covaries with response during on-drug versus off-drug periods.

- High- and low-biomarker participants share the same expected drug response but differ in their biomarker-response covariance.
- The interaction signal resides entirely in the second moment and is eroded directly by carryover.
- This is the core DGP in Hendrickson et al. (2020).

Covariance-only is an unusual restriction in the psychometric tradition; its tractability for MVN power simulation is its main appeal, but it lacks a clean natural generative mechanism.

- FMM and GMM parameterizations typically allow class-specific means at minimum; the means-constant-covariance-varies case tests measurement invariance, not primary DGP structure.
- The encoding's appeal is tractability: the entire DGP reduces to a single MVN draw with a constructed covariance.
- Its cost is the absence of a natural mechanistic interpretation.

8.2 Combined mean and covariance moderation

The Arminger mean-and-covariance form is the general psychometric parameterization, subsuming both pure mean moderation and covariance-only as special cases.

- Both the expected-response vector and the residual covariance matrix may differ between classes and depend on covariates.
- Mean-moderation only and covariance-only are limiting cases.
- This is the general form against which both architectures A and B are constrained submodels.

Under combined moderation, the mean channel is robust to carryover and the covariance channel is eroded, predicting intermediate power loss and making this the most biologically plausible regime.

- Both channels carry class-membership information; the mean channel via $E[BR | B]$, the covariance channel via $\text{Var}[BR | B]$.
- Carryover attenuates the covariance channel but not the mean channel, yielding intermediate power loss.

- The combined specification is plausibly the most biologically faithful regime for biomarker-guided psychiatric trials.

8.3 Location-scale moderation

GAMLSS-style distributional regression allows the biomarker to enter the mean, variance, and higher distributional moments simultaneously via separate link functions.

- Standard regression models only the conditional mean; GAMLSS models the mean, variance, skewness, and kurtosis each as a separate function of covariates.
- The familiar special case is heteroscedastic linear regression with covariate-dependent residual variance.
- GAMLSS generalizes this to arbitrary distributional shape.

GAMLSS occupies an intermediate position between the mean-moderation architecture and B, allowing joint mean and scale moderation without discrete class structure.

- A GAMLSS DGP allows a shifted mean (the mean-moderation architecture location effect) and altered variability (the covariance-moderation architecture analogue) jointly.
- It is the continuous-heterogeneity analogue of FMM, combining mean and scale effects without requiring class structure.
- The intermediate position is attractive for sensitivity analysis as it avoids committing to a class count.

8.4 Heterogeneous random-effects models

Heterogeneous random-effects models replace the single normal random-slope distribution with a mixture, so responder and non-responder classes have distinct treatment-slope distributions.

- In the standard LMM the random treatment slope is drawn from a single normal; the heterogeneous extension replaces it with a finite mixture.
- Responders draw from a large-slope component; non-responders from a near-zero-slope component.
- The biomarker predicts which mixture component a participant comes from.

1cmm is the closest available tooling for heterogeneous random-effects N-of-1 analyses, fitting a mixture on the random-effects distribution and returning posterior class probabilities.

- The framework represents responder heterogeneity at the participant-specific random treatment effect – the most clinically transparent representation.
- The biomarker predicts class membership rather than entering the fixed-effect mean structure.

- `lcmm` accepts standard LMM syntax and is the lowest-friction extension of the existing simulation framework.

9 Implications for analysis and design

The covariance-only DGP is the most restrictive case of a richer family; sensitivity analyses based on it alone risk overstating carryover-induced power loss.

- Covariance-only places the entire biomarker-outcome signal in a channel eroded by carryover by construction.
- Mean-structure signal survives carryover, attenuating overall power loss under combined or mean-only specifications.
- A complete sensitivity analysis should span pure mean moderation, pure covariance moderation, and a combined specification.

Identifiability concerns bound mixture-based analysis to a sensitivity and DGP role at N-of-1 sample sizes, not a primary analysis role.

- Published FMM applications involve hundreds to thousands of participants; N-of-1 trials at $N \in [30, 150]$ are an order of magnitude smaller.
- Mixture models should not be relied on as the primary analysis at trial-relevant N .
- Mixture analysis retains value as a sensitivity check and as a DGP stress-test for the primary linear-mixed analysis.

The heterogeneous random-slopes form is the natural DGP extension target: it admits latent-class generative structure while preserving `nlme::lme` analysis-model compatibility.

- Random-slopes parameterization extends the DGP in a direction internal to the standard mixed-effects framework.
- The analysis model remains the pre-specified linear mixed-effects fit, preserving the audit chain.
- Direct comparison against the covariance-only baseline is possible under matched effect-size calibration.

10 Software for mixture analyses

Mplus is the reference implementation for GMM, FMM, and mixture-SEM; direct comparison with the identifiability literature occasionally requires access to it.

- Mplus is the standard environment for the psychometric mixture-model simulation literature on identifiability and overextraction.
- The Bauer-Curran, Lubke-Muthen, and Nylund studies all use Mplus.
- Reproducing published benchmarks may require Mplus access alongside the R implementations.

11 Discussion

The biomarker-moderation problem sits at the intersection of biostatistical and psychometric traditions that have not previously engaged; recognizing the encoding choice opens both sensitivity-analysis and analysis-strategy programs.

- Biostatistics has efficient within-participant estimators but encodes moderation exclusively as a continuous interaction.
- Psychometrics has mature finite-mixture machinery applied only to cross-sectional settings well above the N-of-1 range.
- The continuous-versus-discrete choice is a substantive scientific question, not a parameterization convenience.

Two limitations bound what the paper currently provides: the five-study simulation program has not been executed, and the prazosin-PTSD re-analysis has not been carried out.

- The companion simulation plan describes but does not execute the empirical core of the present line of work.
- The prazosin-PTSD dataset is the most immediate real-data application target.
- Re-analysis should span at least three biomarker-moderation encodings and a class-aware analysis where convergence permits.

Three open questions remain: primary usability of class-aware analyses at small N , analysis half-life specification under a mixture DGP, and whether trial design should respond to the encoding question.

- Whether class-aware analyses are usable as primary tests or only as sensitivity checks requires direct simulation in the trial-relevant regime.
- The continuous-decay $D_{bc,it}$ has a clean single-component interpretation; its meaning under class-specific carryover dynamics is less obvious.
- If a latent-class DGP is biologically more plausible, designs minimizing carryover may gain power the standard hybrid design surrenders.

The five-study simulation program addresses the open questions sequentially, with pre-registered ADEMP designs, uniform analysis wrappers, and scoped infrastructure additions.

- Factorial designs are pre-registered through ADEMP tables committed before production runs.
- Class-aware analyses are wrapped uniformly for direct cross-study comparability.
- The required DGP infrastructure additions are enumerated, scoped, and sequenced to support both this paper and the companion program.

11.1 Why the analysis model is simpler than the data-generating process

The asymmetry between a structured DGP and a simple four-term analysis model is intentional: a component-matched analysis is impractical at trial-relevant N .

- The DGP decomposes into BR, PB, and TV components; the analysis model is $Sx \sim \text{bm} + \text{t} + \text{Dbc} + \text{bm}:\text{Dbc}$.
- A component-by-component mirror would inflate the variance of $\beta_{\text{bm}:D}$ at $N \in [30, 150]$.
- Four reasons are given below for why the simpler analysis is preferred.

Component-matched analysis is impractical at trial-relevant N for four reasons: identification requirements, parameter inflation, convergence fragility, and robustness of the moderation estimand.

- Many designs supply only a subset of the contrasts needed to identify BR, PB, and TV components separately.
- Each Gompertz component adds 3–4 population parameters, inflating $\text{Var}(\hat{\beta}_{\text{bm}:D})$ at $N \in [30, 150]$.
- $\beta_{\text{bm}:D}$ is robust to BR-versus-PB attribution, making the simpler model adequate for the validation question.

Adding a phase indicator is the minimal tractable extension that tests whether the apparent moderation shrinks under blinding.

- The phase-indicator extension tests whether the drug effect and its moderation shrink under blinding – a PB-mediation signature.
- For papers where the primary estimand is the marginal moderation, the simple four-term model is adequate.
- The phase-indicator extension is a candidate sensitivity analysis for Study 1 once the primary cells are run.

11.2 The latent-class DGP as a generator of apparent subadditivity

The latent-class DGP is a natural generator of apparent subadditivity (a combined response smaller than the sum of the separate component responses) in a one-component analysis; the $\gamma > 0$ finding is ambiguous between direct mechanistic interaction and emergent class structure.

- No direct $BR \times PB$ interaction is required in the class-conditional model to generate a non-zero $\hat{\gamma}$.
- Two mechanisms produce $\gamma > 0$: direct mechanistic coupling and emergent covariance from class structure.
- The companion BR-PB-TV decomposition paper defines and tests both mechanisms.

Emergent subadditivity arises when class membership correlates

with both BR and PB strength, producing a non-zero marginal $E[BR \cdot PB]$ covariance term that mimics a mechanistic interaction.

- Responder-class participants having higher BR and PB on average induces a positive $E[BR \cdot PB]$ covariance via the unobserved class label.
- The magnitude depends on the gating function $\pi(B)$.
- From the additive-extension perspective this is indistinguishable from a γ -weighted mechanistic interaction.

The Study 1 `pb_class_correlation` axis lets the analyst distinguish emergent from mechanistic subadditivity empirically within the simulation output.

- The axis crosses PB-class-independent and PB-class-correlated cells at matched marginal effect size.
- Large $\hat{\gamma}$ in the PB-class-correlated cell but small in the PB-class-independent cell identifies the emergent mechanism.
- The cross-paper synthesis with the BR-PB-TV decomposition paper directly falsifies or confirms class-aware versus mechanistic interpretations of apparent subadditivity.

11.3 Progress to date

Phase 1 has committed three infrastructure components: ADEMP pre-registration, the `lcmm_analysis()` wrapper, and the heterogeneous-random-slopes DGP.

- ADEMP tables cover approximately 2,300 cells across the five studies with seeded RNG control and a deviation-log specification.
- The class-aware analysis wrapper uses a two-stage fit anchoring grid-search starts from a single-class `hlme`.
- The heterogeneous-random-slopes DGP adds a Bernoulli class- membership channel with logistic gating in the standardized biomarker.

The `lcmm_analysis()` wrapper extracts two moderation signals: the primary gating-coefficient z-test and the secondary within-class $\beta_{bm:D}$ t-test.

- The two-stage fit anchors gridsearch starting values from a single-class `hlme` before fitting `hlme(ng=2)`.
- The gating-coefficient z-test identifies whether the biomarker predicts class membership.
- The within-class $\beta_{bm:D}$ t-test is retained for inferential symmetry with the linear-mixed baseline.

Smoke-test results confirm the heterogeneous-random-slopes DGP produces the expected class structure and that `lcmm` discriminates it from the single-component MVN-DGP.

- Single-replicate test at $N = 70$, gating slope 1.5 produces 46% responders and marginal $\text{Cor}(B, BR) = 0.63$.

- The linear-mixed `bm:Dbc` test detects the signal ($p = 0.002$) without class awareness.
- `lcmm` returns entropy 0.86 and BIC 170 lower than the matched single-component fit.

The 100-replicate Study 5 pre-flight pilot establishes a 32:1 per-replicate cost ratio for the class-aware versus linear-mixed analysis, with 96% `lcmm` convergence.

- Per-replicate timing: 0.05 s linear-mixed baseline; 1.57 s class-aware analysis.
- `lcmm` convergence rate: 96%; four failures arose from gridsearch initialization.
- Results are checkpointed at `output/03-study5-pilot.rds`.

The unconditional Wald-on-gating inflation is a finite-sample identifiability artifact: the conditional Wald SE underestimates true variability because the optimizer latches onto random class boundaries.

- Empirical SD of the gating coefficient (63.9) is fifty times the median within-replicate Wald SE (0.86).
- The gating coefficient is centered at zero with no systematic bias; inflation is in the SE, not the estimate.
- This is the overextraction artifact predicted by Bauer and Curran (2003) and Lubke and Muthén (2005).

The naive likelihood-ratio test (LRT) is anti-conservative because the true distribution at the $ng = 1$ boundary is chi-bar-squared, not chi-squared; the bootstrapped LRT is correct but too expensive.

- Rejection rate 0.146 under the naive chi-squared($df = 4$) reference.
- The true LRT distribution at the $ng = 1$ boundary is a chi-bar-squared mixture (McLachlan and Peel, ch. 6).
- The Nylund et al. (2007) bootstrapped LRT is the standard calibration but is too expensive at the class-aware simulation per-fit budget.

The BIC-conditional gating procedure controls Type I at $N = 70$ under the null: BIC selects $ng = 2$ in 1% of replicates and none of those also reject the gating Wald.

- The procedure requires $\Delta_{\text{BIC}} > 0$ (or > 6 per Raftery 1995) before testing the gating coefficient.
- Zero conditional procedure rejections at the null gives Type I = 0 at the pilot's resolution.
- Whether the procedure has adequate power under a true two-class DGP is the central question for the rest of Study 5.

The earlier claim that class-aware analyses are not usable as primary tests at small N requires correction: it applies to the unconditional Wald test, not to the BIC-conditional procedure.

- The corrected proposition is that a single mixture-model coefficient's Wald test is unreliable at small N .
- A model-comparison wrapper (BIC or BLRT) is required to control Type I error.
- The BIC-conditional procedure is the pragmatic primary test at trial-relevant N .

The full five-study program is feasible on a single workstation in approximately one week of background compute at 8–16 cores.

- Study 1's class-aware arm requires approximately 340 serial hours, or 22–44 hours on 8–16 cores.
- Study 5 production is the largest block at approximately 60 hours per arm on 8 cores.
- The calibration sub-study and Study 5 pre-flight should run first to finalise effect-size axes before committing to Studies 1, 2, and 3.

The open question raised by the pilot is whether the BIC selection step has sufficient sensitivity under true class structure: Pilot 4 will test this across gating slopes and class-separation values.

- A conservative null-controlling procedure that rarely fires under true class structure provides no value.
- Expected pattern: BIC-conditional outperforms lme at large class separations; underperforms at moderate separations.
- Pilot 4 will fit the BIC-conditional procedure to the heterogeneous-slopes DGP from Study 3.

11.4 Pilot results (Study 5, 240 replicates)

The 240-replicate Study 5 pilot crossed three gating-slope levels and three test forms at $N = 70$, with MCSE approximately 0.030 at working power.

- Wall time 558 s on 4-core `parallel::mclapply`.
- `lcmm` convergence: 92.9% at the null, 98.3–98.8% at non-null cells.
- MCSE approximately 0.030 at working power near 0.5.

The unconditional Wald inflates Type I to 0.197; the BIC-conditional procedure controls Type I but is severely under-powered, approximating a never-reject rule under heterogeneous slopes.

- Unconditional Wald Type I 0.197 (95% CI [0.145, 0.250]), decisively above the nominal 0.05; power below the lme baseline at both non-null cells.
- BIC-conditional Type I 0.004 under the null; power 0.025 at $gs = 1.0$ and 0.004 at $gs = 1.5$.
- Non-monotonicity at $gs = 1.5$ suggests the lme captures the heterogeneous-slopes signal well enough that BIC continues to favor $ng = 1$.

The lme bm:Dbc test dominates both lcmm tests at this N and DGP regime; a canonical 1,000-replicate run is required to map the regime where the BIC-conditional procedure becomes competitive.

- lme holds nominal Type I (0 of 240) and reaches 0.542 power at $gs = 1.5$.
- Neither class-aware test is competitive: the unconditional test is anti-conservative; the BIC-conditional test is severely under-powered.
- A canonical run at $N \in \{35, 70, 100\}$ with a wider gating-slope grid is required to locate the competitive regime, plausibly at large class separations.